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THAPAR INSTITUTE OF ENGINEERING & TECHNOLOGY, PATIALA

Department of Computer Science & Engineering

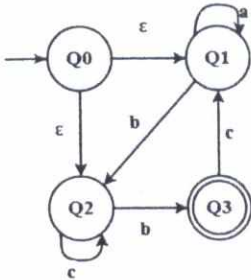
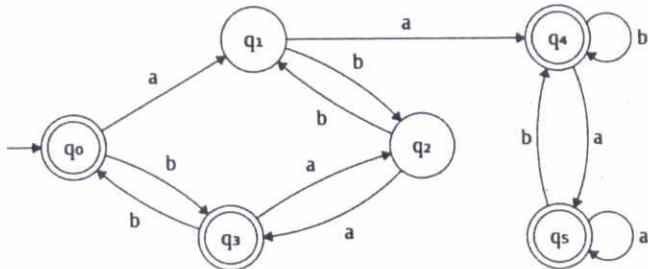
Theory of Computation (UCS701) – Mid Semester Examination

Date: 01/03/2025 & Time: 3:00PM

MM: 30 & MT: 120 Min

Faculty Name: Dr. Chinmaya Panigrahy, Dr. Shashank Sheshar Singh, Dr. Rohan Sharma, Dr. Aditi Sharma, Dr. Vibha Jain, Dr. Nidhi Chakravarty, Mr. G.V. Eswara Rao, and Mr. Abdul Kadir

Attempt all the questions. Attempt/Answer all sub-parts like (a), (b), (c) for each question in one place. Do mention Page No. of your attempt on the front page. Question paper is of two pages. Please turn over this page to see the questions on the next page.

Ques. No	Questions	Mark	CO	BL
Q1.	(a) Apply Thompson's construction to construct an NFA with epsilon moves for the regular expression, $(aa+b)^*(bb+a)^*$.	2	CO1	L3
	(b) Design a DFA that accepts all the binary strings divisible by 3.	2	CO1	L3
	(c) Construct a Moore machine that takes a binary number as input and produces residue modulo 5 as output. Convert the resultant Moore machine to an equivalent Mealy machine.	3	CO1	L3
	(d) List all the minimal length strings not in the language represented by the regular expression, $1^*(01)^*0^*$.	1	CO1	L2
Q2.	(a) Convert the given NFA with epsilon moves to an equivalent DFA. 	3	CO1	L3
	(b) Prove that the regular languages are closed under set-difference operation.	1.5	CO1	L3
	(c) Minimize the given DFA. 	2.5	CO1	L3
Q3.	(a) Prove that the language $L = \{ww^R : w \text{ is a string in } \Sigma^* \text{ and } w^R \text{ is the reverse of } w\}$ is not regular using Pumping lemma for regular languages. Consider, $\Sigma = \{a,b\}$.	2.5	CO1	L4
	(b) Construct an NFA from the regular expression, $(a+b)^*aab$, and convert the NFA into a resultant DFA.	2	CO1	L4

	(c)	Find the regular expression corresponding to the given finite automata using Arden's theorem.	<pre> graph LR start(()) --> q1((q1)) q1 -- b --> q2((q2)) q2 -- a --> q1 q2 -- a --> q3(((q3))) q3 -- b --> q2 q1 -- a --> q1 </pre>	2.5	CO1	L4
Q4.	(a)	Suppose that you are taking a walk with your dog along a straight-line path. Your dog is on a leash that has a length of two, meaning that the distance between you and your dog can be at most two units. You and your dog start at the same position. Consider the alphabet $\Sigma = \{y, d\}$. A string in Σ^* can be thought of as a series of events in which either you (y) or your dog (d) moves forward one unit. For example, the string “yydd” means that you take two steps forward, then your dog takes two steps forward. Let $L = \{w \in \Sigma^* \mid w \text{ describes a series of steps that ensures that you and your dog are never more than two units apart}\}$. For example, “yydd” and “yyddydd” are valid walks, while “ddydd”, “yydyyd”, “yyydd”, and “yddyyddd” are invalid. Construct a minimal DFA for L.	3	CO4	L3	
	(b)	Let $M = (Q, \Sigma, \delta, q_0, \{q_f\})$ be a ϵ -NFA such that there are no transitions into q_0 and there are no transitions out of q_f . Specify the language accepted by each of the following machines in terms of L, where L is the language accepted by the machine M. You must provide a brief justification for your answer. (i) The machine M1 obtained from M by adding a ϵ-move from q_f to q_0. (ii) The machine M3 obtained from M by making every state as final state. (iii) The machine M2 obtained from M by adding a ϵ-move from q_0 to every state that is reachable from q_0.	3	CO4	L3	
	(c)	Write a regular expression to represent all the strings over $\{0,1,2\}$ that does not contain two consecutive 1's.	2	CO1	L3	